

Mohan Konugolu

Robotics Engineer

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SUMMARY

Robotics engineer specializing in reinforcement learning and embodied AI — humanoid locomotion and whole-body control, dexterous manipulation, and sim-to-real training in NVIDIA Isaac Sim, Isaac Lab, and Isaac Gym. Experienced with imitation learning, SMPL-X motion retargeting, PPO / SKRL training, and foundation-model perception (Isaac ROS, FoundationPose), with a focus on vision-language-action (VLA) and foundation-model robotics. Strong Python / PyTorch and ROS 2 engineering background, with every project containerized in Docker for reproducible deployment.

TECHNICAL SKILLS

Reinforcement & Imitation Learning: PPO, SKRL, Isaac Lab, Isaac Gym, reward shaping, sim-to-real transfer, motion retargeting (SMPL-X, PHC)

Embodied AI & Humanoids: whole-body control (HOVER), bipedal locomotion, dexterous manipulation, Unitree G1, five-fingered hands; vision-language-action (VLA) and foundation-model robotics

Foundation Models & Perception: FoundationPose (6-DoF pose), Isaac ROS, OpenCV, YOLO, MiDaS, point-cloud processing

Simulation & Digital Twins: NVIDIA Isaac Sim, Isaac Lab, Isaac Gym, Omniverse USD, Gazebo

Deep Learning: PyTorch, CNNs (MobileNetV3), time-series models (MiniRocket)

Robotics & Middleware: ROS, ROS 2, Nav2; motion planning (pure pursuit, TEB, clothoid)

Sensors & Hardware: 2D/3D LiDAR, solid-state LiDAR, IMU, RGB-D cameras, force/torque sensors; AMRs, manipulators, humanoids

Languages & Deployment: Python, C++, Bash; Docker (all projects containerized), Git, Linux

KEY PROJECTS

Custom Humanoid Locomotion (Hyperspawn) — *Isaac Sim · Isaac Lab · PyTorch · PPO*

- Imported an open-source Hyperspawn humanoid featuring 4-bar linkage mechanisms and a Stewart (parallel) platform, solving the closed-chain kinematics in Isaac Sim so simulated dynamics matched the physical robot.
- Trained a bipedal walking policy with reinforcement learning and integrated an SMPL-X motion-retargeting pipeline (PHC) for combined imitation + RL; credited as a contributor to the Hyperspawn open-source project.

Humanoid Whole-Body Gesture Control (Unitree G1) — *HOVER · SMPL-X · Isaac Lab*

- Trained the Unitree G1 humanoid to perform naturalistic, human-like conversational gestures using the HOVER whole-body-control framework.
- Adapted the motion-retargeting pipeline to ingest SMPL-X data and retarget human motion onto the G1, then extended the task to five-fingered hands for lifelike hand movement during conversation.

Dexterous Brick-Laying Manipulation — *Isaac Sim · Isaac Lab · SKRL · PPO*

- Trained a five-fingered industrial manipulator to approach with an open hand, grasp a brick with all five fingers, lift, and place it at a target using PPO (SKRL).
- Instrumented fingertips with force sensors and added IMUs for closed-loop, contact-aware grasping in a custom Isaac Sim environment.

RL Motion Control for Autonomous Navigation — *Isaac Sim · ROS 2 · Nav2*

- Integrated a reinforcement-learning motion-control policy with the Nav2 stack plus custom low-speed smoothing for stable steering.
- Built a custom-vehicle Isaac Sim scene (LiDAR, camera, IMU, motor control) bridged to ROS 2; trained and tested policies that avoid slow-moving pedestrians while navigating point to point.

Robot Factory Digital Twin — *Isaac Sim · ROS 2 · Isaac Lab*

- Built physics-accurate Isaac Sim digital twins of robot factories with multiple coordinated robots and simulated LiDAR, camera, and IMU streams.
- Wired the twin to ROS 2 and Isaac Lab so client teams could validate control logic and train RL policies before hardware deployment.

Foundation-Model Perception: 6-DoF Pose Pipeline — *NVIDIA Isaac ROS · FoundationPose*

- Implemented a FoundationPose 6-DoF object pose-estimation and tracking pipeline using NVIDIA Isaac ROS.
- Integrated the pose output into Isaac Sim and a ROS 2 perception graph for a client perception system.

Autonomous Mobile Robot Software Stack — *C++ · ROS · 2D LiDAR*

- Built a full AMR autonomy stack: double-look-ahead pure-pursuit tracking, clothoid path generation, and TEB local planning.
- Coded 3-zone 2D-LiDAR safety monitoring and a trolley-leg detection algorithm for autonomous docking.

Monocular Depth Estimation — *PyTorch · YOLO · MiDaS*

- Trained a deep network (YOLO boxes + MiDaS relative depth) for absolute object-level depth — 0.18 m MAE in-house, 1.1 m on nuScenes; selected for a poster at Bosch Germany's internal AAI research conference.

LiDAR Cover-Glass Blockage Detection — *PyTorch · MiniRocket · MobileNetV3*

- Detected cover-glass occlusion on a solid-state LiDAR with MiniRocket (photon time-series) and MobileNetV3 (3-channel point-cloud images), both above 90% accuracy.

PROFESSIONAL EXPERIENCE

Robotics Consultant (Freelance) — Upwork Feb 2024 – Present

Remote

- Top Rated consultant (100% Job Success, 20 jobs, 597+ hours) training RL policies and building Isaac Sim digital twins for robotics companies worldwide; every project containerized with Docker.
- Specialize in humanoid and manipulator reinforcement learning (Isaac Lab, SKRL, PPO), motion retargeting (SMPL-X), and foundation-model perception (Isaac ROS, FoundationPose).
- Build Isaac Sim digital twins of factories and custom robots (AMRs and manipulators) and port full software stacks to ROS 2 / Isaac Lab for control and RL.

Software Developer — Bosch Global Software Technologies (BGSW) Sep 2022 – Feb 2024

India

- Built reinforcement-learning vehicle control and computer-vision depth perception for autonomous driving, validated on in-house test vehicles.
- Developed ML occlusion detection for an in-house solid-state-LiDAR cover glass (above 90% accuracy).

Robotics Engineer — ZeroPoint Robotics Aug 2021 – Aug 2022

India

- Core engineer on an autonomous mobile robot; built pure-pursuit path tracking, clothoid path generation, TEB local planning, and 2D-LiDAR safety.
- Implemented 3-zone 2D-LiDAR safety and a trolley-leg detection algorithm for autonomous docking.

Deployment Engineer — THRSL Dec 2020 – Jul 2021

India

- Built pallet-detection perception and deployed the autonomous mobile robot on-site.
- Commissioned two autonomous pallet jacks in a live factory: 2D-LiDAR mapping, localization, safety functions, and extended long-term mission testing.

EDUCATION

MSc, Robotics and Artificial Intelligence Sep 2024 – Sep 2025

Cardiff Metropolitan University — Cardiff, UK

BEng, Mechatronics 2016 – 2020

Chandigarh University — Punjab, India